

BENEDIKT SEITER

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EDUCATION

ETH Zürich

Zurich, Switzerland

M.Sc. in Applied Mathematics, focus area: Machine Learning

Sep 2026–Spring 2028 (expected)

Current courses: Probabilistic Artificial Intelligence; Natural Language Processing; Advanced Machine Learning.

B.Sc. in Mathematics

Sep 2023–Aug 2026

GPA: 5.6/6.0 (1 SD above the graduating cohort mean)

Thesis: *Anchor PCA* (6.0/6.0), advised by Prof. Jonas Peters.

Key courses: Mathematical Statistics; High-Dimensional Statistics; Computational Statistics; Machine Learning.

RESEARCH INTERESTS

Domain generalization; foundation models; causality; mechanistic interpretability; LLM evaluation; post-training.

PUBLICATIONS

B. Seiter, A. Fries, J. von Kügelgen, and J. Peters. *Anchor PCA*. *NeurIPS 2026* (accepted). [arXiv:2606.06233](https://arxiv.org/abs/2606.06233).

RESEARCH & PROJECTS

Anchor PCA

Apr 2026–present

ETH Zürich, Seminar for Statistics

[PDF](#) [GitHub](#)

- Developed Anchor PCA for multi-domain dimensionality reduction, balancing explained variance with stability under domain shift; implemented estimators in Python with NumPy and SciPy.
- Benchmarked against pooled and worst-case PCA in an end-to-end pipeline; improved out-of-domain explained variance on financial and gas sensor data.
- Established recovery of a maximal invariant subspace and proved minimax robustness guarantees.

Transformer Interpretability and Experimentation

Ongoing

Independent study through [ARENA](#); implementing activation interventions, linear probing, and sparse autoencoders in PyTorch; running controlled experiments on open-weight LMs using remote GPUs.

Benchmarking Invariant Risk Minimization for Financial Time Series

Sep 2025–Jan 2026

ETH Zürich, Supervised Research Project

Reproduced and extended InvariantStock, an IRM-based VAE for regime-robust equity prediction, in PyTorch; rebuilt its pipeline and benchmarked VIX-based environments and loss variants.

Seminar on Representation Learning

Sep 2025–Jan 2026

ETH Zürich

Analyzed and presented papers on deep learning, autoencoders, ICA, contrastive and self-supervised learning, causality, and single-cell biology applications.

TEACHING

Teaching Assistant

Sep 2025–present

ETH Zürich

Lead weekly problem-solving sessions in Theoretical Computer Science and Probability & Statistics.

AWARDS

First Prize, Bundeswettbewerb Mathematik

2023

Top 2.5% in Germany's national mathematics competition; invited to the German IMO team selection examinations.

DPG-Abitur- und Buchpreis (Physics)

2023

Dr. Hans Riegel-Fachpreis (Mathematics)

2022

SKILLS

Programming: Python, C++, R, MATLAB **Libraries:** PyTorch, NumPy, SciPy, scikit-learn, pandas

Tools: Git, LaTeX **Languages:** German (native), English (C2, IELTS 8.5)